

Class 11:

1. A lamina is bounded by the curves  $y = x^2 - 3x$  and  $y = 2x$ . If the density at any point is given by  $\lambda xy$ , find by double integration, mass of the lamina. Ans:  $182\frac{7}{24}\lambda$
2. Find the mass of the lamina in the form of the cardioid  $r = a(1 + \cos \theta)$  whose density at any point varies as the square of its distance from the initial line. ans:  $\frac{21\pi a\mu^4}{32}$
3. Find the mass of a solid in the form of the positive octant of the sphere  $x^2 + y^2 + z^2 = 9$  if the density at any point is  $2xyz$  ans: 30.375
4. Find the centroid of the area enclosed by the parabola  $y^2 = 4ax$ , the axis of x and its latus rectum. Ans:  $(\frac{3a}{20}, \frac{3a}{16})$

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